Flow Matching

Discussion #10

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1 Warmup: Noise Shapes

At their core, diffusion and flow matching models sample from a base probability *distribution* (usually Gaussian noise) and transport it to a target data distribution.

Note: In this sheet we use the “flow” formulation. Don’t confuse it with the “diffusion” one!

For each task below, state the shape of the input noise x_0 (a.k.a. x_{noise}).

0) Generating a grayscale, 1920×1080 image

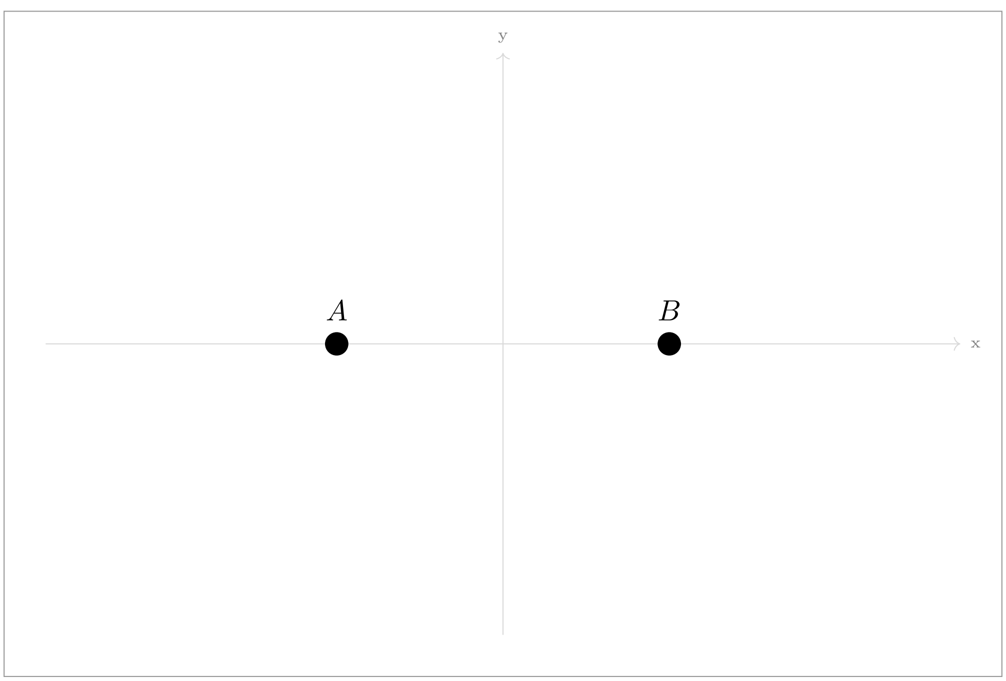
$(1, 1080, 1920)$ or $(1080, 1920)$ flattened to $(1920 \times 1080,)$

1) Generating a random point (x, y) within a square

$(2,)$ — just the x and y coordinates.

2 Visualizing Flow in 2D

We want to train a flow matching model to generate samples from a data distribution consisting of exactly two points, $A = (-1, 0)$ and $B = (1, 0)$ (50% probability each).



1.0) Define the noise region. Sketch in the plot above the contour of $\mathcal{N}(0,1)$ at 1-std deviation (to roughly denote the high probability region of the noise).

Conditional Flow During each iteration of diffusion model training, we select a data point d (either A or B) at random and combine it with random noise r according to timestep $t \in [0,1]$ like $x_t = d \cdot t + (1-t) \cdot r$.

1.1 Pick **one** "random" r (full noise) and plot them in the chart.

1.2 For each of these random samples generate x_0 (noisy) from A and B respectively and **plot** them on the chart. Do the same for $x_{0.9}$ (almost clean).

1.3 One popular method of supervising flow models is to use the velocity vector which we use to guide model outputs: ((x_1-x_0) - $u_\theta(x_t, t)$). For each x_t **draw** the corresponding velocity vector (don't worry about the length).

Marginal Flow Our diffusion model models marginal flow, which for a given x_t is the **weighted average** over all conditional flows from the datapoints we train on.

1.4 For the point r, **plot** marginal flows for $t = 0$ and $t = 0.9$ (roughly).

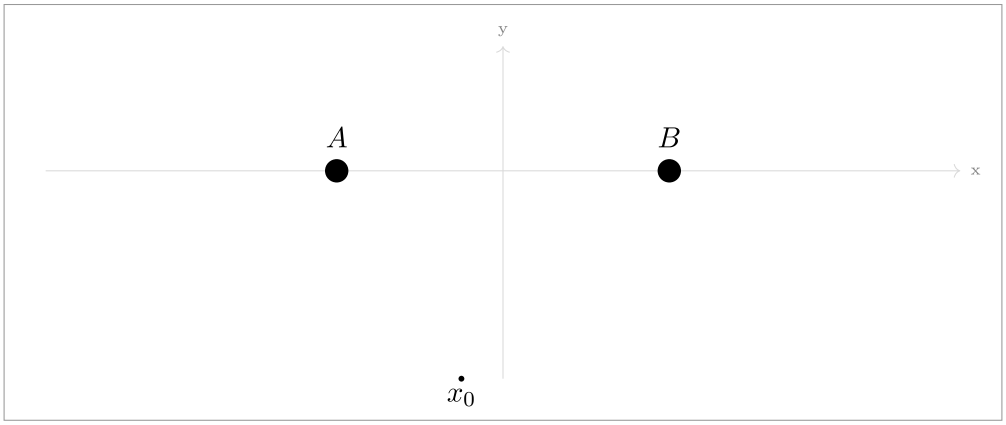
1.5 In English, come up with a general rule for where the marginal flow points towards at $t = 0$ (full noise) for any type of data.

At full noise, the flow will always point towards the average (centroid) of the data distribution. Intuitively, for every noise point, during training they have equal probability of being supervised towards any of the data, so the best possible prediction to make is to predict the average (will which minimize mean error).

1.6 In english, come up with a general rule for where the marginal flow points towards at $t = 0.9$ (almost clean) for any type of data.

Almost to the nearest neighbor in the dataset.

1.7 In the plot below for the given x_0, simulate the diffusion model sampling procedure using 4 sampling steps, drawing the intermediate x_t.

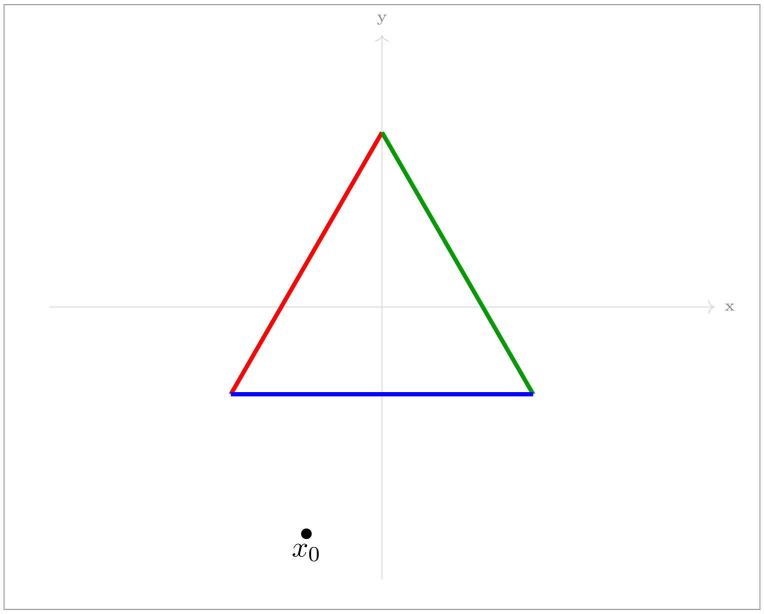


1.8 What if I wanted to generate samples of point A twice as frequently as point B? How could I modify the standard training pipeline to achieve this?

> During training, sample A twice as frequently as B! or, define a third datapoint C at the same position as A.

3 Conditioning and Guidance

Now suppose we want to draw from a distribution corresponding to sides of a triangle (think of this like our little image manifold). In addition, we want to *control* what type of point our model generates: red, green, or blue. (not the color, but the location)



3.1 Adding control. The most common way to control diffusion models is by additional *conditioning* input, like language, to the model.

Explain how one could sample from a specific color without additional model input by changing training.

Train one model per color. We will have: Red model, Green model, and Blue model.

Classifier-Free Guidance (CFG) modifies the vector field to nudge samples more towards red compared to an unconditional sample:

$$\tilde{u_\theta}(x, t, c) = u_\theta(x, t, \varnothing) + w \cdot (u_\theta(x, t, y) - u_\theta(x, t, \varnothing))$$

where u_θ is our neural network model, which learns marginal flow.

3.2 Prompted Flow: Draw in the diagram where $u_\theta(x, t, \textit{red})$ and $u_\theta(x, t, \textit{green})$ point.

Each will point towards the center of the red and green lines respectively (because of the same effect as 1.5. This is because we're at full noise!)

3.3 Unprompted Flow: Draw in the diagram where $v(x_0, t, \varnothing)$ points

It will point to the center of the triangle (center of *all* the data).

3.4 Guidance: Draw the CFG vector $\tilde{u_\theta}$ for a large w. How does it differ from the standard prompted vector?

With a high w, it will point more and more away from the center of the triangle and more towards the side of the triangle.